

Mitochondrial DNA Intraspecific Variability in *Sergentomyia minuta* (Diptera: Psychodidae)

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J. Med. Entomol. 1–10 (2015); DOI: 10.1093/jme/tjv075

ABSTRACT Recently, there has been growing interest in analysis of the geographical variation between populations of different *Phlebotomus* spp. and American sand flies by comparing the sequences of various genes. However, little is known about the genetic structure of the genus *Sergentomyia* França & Parrot. No study has been carried out on *Sergentomyia minuta* Rondani. Most authors recognize this as a species with a high degree of morphological polymorphism, and some suspect that there are two subspecies: *Se. minuta minuta* Rondani in Europe, having about 40 horizontal cibarial teeth (sticks aligned along a straight line in the cibarial cavity), and *Se. minuta parroti* Adler & Theodor in North Africa, having about 70 cibarial teeth. Here we analyzed phylogeographic patterns using cytochrome b (Cytb) and cytochrome C oxidase I mtDNA for 29 populations from 10 countries: Algeria, Cyprus, France (continental and Corsica), Greece (continental and Crete), Malta, Montenegro, Morocco, Portugal (continental and Atlantic Savage Islands), Spain, and Tunisia. We analyzed intra- and interpopulation patterns of genetic diversity. Our results from Bayesian inference showed a complex genetic structure of *Se. minuta* with four haplogroups including many different haplotypes. One haplogroup includes all the specimens from North Africa. A second haplogroup includes a few specimens from the south of France, Spain, and one from Portugal. The third includes many specimens from southern France, all the specimens from Corsica, one from Spain, and all specimen from Portugal except one. A fourth branch includes specimens from the Balkans, Malta, Crete, Cyprus, and curiously some from the Atlantic Savage Islands; settlement of the latter population remains unexplained. However, our results suggest that the settlement of the Mediterranean basin could have occurred at the same time for *Se. minuta* and both *Phlebotomus perniciosus* Newstead and *Phlebotomus ariasi* Tonnoir. The spatial distribution of haplotypes was congruent with phylogenetic findings.

RÉSUMÉ Récemment, un intérêt grandissant s'est manifesté pour l'analyse des variations géographiques entre populations de différentes espèces du genre *Phlebotomus* Rondani et de Phlébotomes américains par comparaison de marqueurs moléculaires variés. Cependant, la structure génétique du genre *Sergentomyia* França & Parrot demeure mal connue et aucune étude n'a été conduite chez *Sergentomyia minuta* Rondani dont le statut taxinomique est discuté. La majorité des auteurs considère cette espèce avec un degré élevé de polymorphisme tandis que certains autres considèrent deux sous-espèces : *Se. minuta minuta* Rondani en Europe, avec environ 40 dents cibariales et *Se. minuta parroti* Rondani en Afrique du nord avec environ 70 dents cibariales. Nous avons mené une étude phylogéographique basée sur deux marqueurs mitochondriaux : le cytochrome b et le cytochrome C oxydase I chez 29 populations provenant de 10 pays (Algérie, Chypre, Espagne, France (continentale et Corse), Grèce (continentale et Crète), Malte, Maroc, Monténégro, Portugal (continental et îles sauvages) et Tunisie). Nous avons également analysé les variations intra- et inter-populationnelles. Nos résultats montrent une structure génétique complexe chez *Se. minuta* avec quatre branches qui incluent un grand nombre d'haplotypes. L'une d'elles inclut tous les exemplaires d'Afrique du nord. Une seconde branche inclut quelques spécimens de sud de la France, Espagne et un spécimen de Portugal. La troisième branche inclut

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plusieurs spécimens de France, tous les spécimens de Corse, un spécimen d'Espagne et presque tous les spécimens du Portugal. La quatrième branche inclut les spécimens des Balkans, de Malte, de Crète, de Chypre, et curieusement ceux des îles sauvages situées dans l'Océan Atlantique dont nous ne pouvons pas expliquer le peuplement. Cependant, nos résultats suggèrent que le peuplement du bassin méditerranéen se serait déroulé de manière contemporaine pour *Se. minuta*, pour *Phlebotomus perniciosus* et pour *Phlebotomus ariasi*. La distribution spatiale des haplotypes identifiés est congruente avec leurs structures génétiques.

KEY WORDS *Sergentomyia minuta*, genetic structure, haplotype, lineage, Mediterranean basin

In medical entomology, there is a dogma that the genus *Sergentomyia* França & Parrot is not involved in the transmission of any pathogen. All over the Mediterranean basin, *Sergentomyia minuta* Rondani is a common and widespread species. Its vectorial role has also been neglected. However, the detection of Toscana virus RNA (Charrel et al. 2006) and of *Leishmania major* Yakimoff & Schokhor DNA (Campino et al. 2013) in this species associated with recent findings about its ability to bite rodents (Jaouadi et al. 2013) questions its vectorial competence, especially if we consider the possible role of *Sergentomyia* spp. in the transmission of *Leishmania infantum* in Senegal (Senghor et al. 2011) and of the role of *Spelaomyia darlingi* Lewis & Kirk in the transmission of *L. major* in Mali (Berdjane-Brouk et al. 2012).

Se. minuta has a doubtful taxonomic status. It is considered by some authors as a unique species having a high intraspecific polymorphism depending on its North–South latitudinal range (Rioux et al. 1975, Léger et al. 1979, Belazzoug et al. 1982, Pesson et al. 1984), whereas several authors recognize two subspecies (Addadi and Dedet 1977, Belazzoug et al. 1982): *Se. minuta minuta* in Europe, having cibarial armature with about 40 teeth, and *Se. minuta parroti* in North Africa, having about 70 cibarial teeth.

To our knowledge, no study based on DNA sequences has been carried out on *Se. minuta*. In the present work, we have used two mitochondrial markers: cytochrome B (Cytb) and cytochrome C oxidase I (COI) to explore the relationships between *Se. minuta* populations. The first marker is commonly used to study phylogenetic relationships within Phlebotomine sand flies, whereas the second one is the most sequenced marker for DNA barcoding programs.

Materials and Methods

Sand Flies Collection. Seventy *Se. minuta* males and females have been processed as indicated in Table 1. They were collected using sticky traps and CDC miniature light traps. They originated from 29 locations from 10 countries (Table 1). Sand flies from France, Greece, and Portugal included populations from the continent as well as from Corsica, Crete, and Savage Islands (Alves-Pires et al. 1997), respectively. All specimens were stored in 70–96% ethanol.

Mounting. The head and genitalia of individual sand flies were cut off in a drop of ethanol, cleared in

boiling Marc-André solution, and mounted between slide and cover slip for identification. The body of each specimen was stored dried in a vial at -20°C waiting DNA extraction and molecular processing. Cibarial teeth were counted using a camera lucida.

DNA Extraction. Genomic DNA was extracted from the thorax, wings, legs, and abdomen of individual sand flies using the QIAmp DNA Mini Kit (Qiagen, Germany) following the manufacturer's instructions, modified by crushing the sand fly tissues by hand using a piston pellet (Treff, Switzerland), and using an elution volume of 200 μl , as detailed in Depaquit et al. (2004).

PCR Amplification and Sequencing. mtDNA amplifications were performed in a 50 μl volume using 5 μl of extracted DNA solution and 50 pmol of each of the primers. The PCR mix contained (final concentrations) 10 mM Tris HCl (pH 8.3), 1.5 mM MgCl_2 , 50 mM KCl, 0.01% Triton X 100, 200 μM dNTP each base, and 1.25 units of Taq polymerase (5 prime, Germany).

PCR of the cytochrome B gene has been done using the primers N1N-PDR: (5'-CA(T/C) ATT CAA CC(A/T) GAA TGA TA-3') and C3B-PDR: (5'-GGT A(C/T)(A/T) TTG CCT CGA (T/A)TT CG(T/A) TAT GA-3') following the method previously published by Essegheir et al. (1997): 5 cycles of (denaturation at 94°C for 30 s, annealing at 40°C for 60 s, and extension at 68°C for 60 s), followed by 35 cycles of (denaturation at 94°C for 60 s, annealing at 44°C for 60 s, and extension at 68°C for 60 s). COI was amplified using the primers designed by Hajibabaei et al. (2006): LepF (5'-ATT CAA CCA ATC ATA AAG ATA TTG G-3') and LepR (5'-TAA ACT TCT GGA TGT CCA AAA AAT CA-3'), under the following thermal profile (Costa et al. 2007): 5 cycles of (denaturation at 94°C for 30 s, annealing at 45°C for 90 s, and extension at 68°C for 60 s), then 35 cycles of (denaturation at 94°C for 30 s, annealing at 51°C for 90 s, and extension at 68°C for 60 s).

Amplicons were analyzed by electrophoresis in 1.5% agarose gel containing ethidium bromide. Direct sequencing in both directions was performed using the primers for DNA amplification. The correction of sequences was done using Pregap and Gap software included in the Staden Package (Bonfield and Staden 1996).

Sequences Analyses. Alignments and nucleotide sequence diversity among the sample were obtained using Mega v6.0 (Tamura et al. 2013), with the Kimura-2 parameter for diversity.

Table 1. Sampling species

Country	Area	City	Population number referring to Figure 1	Catching method	Sex	Specimen	Genbank accession numbers		Haplotype N°	
							CytB	COI	CytB (n = 40 haplotypes)	COI (n = 53 haplotypes)
Algeria	Jijel Aurès	Beni Ahmed Ain Zaatout	1	SP	♀	BEN01	KJ481011	KJ481180	30	43
			2	CDC	♀	SM 530	KJ481017	KJ481120	29	50
	Ghardaia	Mettli	3	CDC	♀	SM 535	KP828501	KP828564	32	46
					♀	SM 605	KJ481015	KJ481118	38	51
					♂	SM 544	KJ481013	KJ481116	3	44
Cyprus	Mila	Vieux Mila Armenochori Agios Giorgios	4	CDC	♀	SM 536	KP828502	KP828565	3	41
					♀	SM 561	KJ481014	KJ481117	39	52
					♀	SM 515	KJ481012	KJ481115	40	53
	Hérault	Roquebrun	13	CDC	♀	SM VIM2	KJ481018	KJ481121	32	49
			22		♀	8Cy	KP828480	KP828544	18	30
France	Alpes-Maritimes	Beaulieu	23	CDC	♀	11Cy	KP828481	KP828545	19	31
					♀	3Fr	KP828462	KP828526	4	14
					♀	4Fr	KP828463	KP828527	10	15
	Puy de Dôme Var	Mezel Le Beausset	24	CDC	♀	5Fr	KP828461	KP828525	12	13
					♀	6Fr	KP828464	KP828528	11	16
France (Corsica)	Corse du Sud	Pianotolli	25	CDC	♀	11Fr	KP828465	KP828529	13	17
					♀	PH 86	KP828466	KP828530	4	10
					♀	PH 89	KP828469	KP828533	14	12
	Sartène	Sartène	26	CDC	♀	PH 90	KP828467	KP828531	11	18
					♀	PH 91	KP828468	KP828532	11	19
Greece (continental)	Macedonia	Petrussia Platania	9	CDC	♀	PH 99	KP828470	KP828534	11	20
					♂	6PIAN	KP828471	KP828535	11	21
					♂	9PIAN	KP828472	KP828536	11	22
	Crete	Skopi	10	CDC	♀	7SM	KP828473	KP828537	11	23
					♀	9SM	KP828475	KP828539	11	34
Greece (Crete)	Macedonia	Petrussia Platania	27	CDC	♀	9-36SM	KP828474	KP828538	15	24
					♀	Dr 4	KP828509	KP828572	36	5
					♂	Dr 14 PM1	KP828506	KP828569	36	5
	Crete	Skopi	11	CDC	♂	Dr 14 PM2	KP828507	KP828570	36	5
					♀	Dr 26	KP828508	KP828571	37	5
Malta	Gozo	Ghajnsielem	12	CDC	♀	4 crete	KP828482	KP828546	21	32
					♀	6 crete	KP828483	KP828547	21	33
					♀	9 crete	KP828485	KP828548	22	34
	Gozo	Ghajnsielem	7	CDC	♀	11 crete	KP828486	KP828549	23	35
					♂	MALT2	KP828487	KP828550	24	36
Gozo	Ghajnsielem	7	CDC	♂	MALT3	KP828490	KP828553	25	38	
				♂	4MALT	KP828489	KP828552	26	38	
				♀	3MALT	KP828488	KP828551	25	37	

Table 1. Continued

Country	Area	City	Population number referring to Figure 1	Catching method	Sex	Specimen	Genbank accession numbers		Haplotype N ^c	
							CytB	COI	CytB (n = 40 haplotypes)	COI (n = 53 haplotypes)
Montenegro	Bar	Kinje	8	CDC	♀	03-48 PF22 03-48 PF24 03-48 PF 28	KP828504 KP828503 KP828505	KP828567 KP828566 KP828568	34 33 35	48 47 47
Morocco	Rif	Chefchaouene	28	CDC	♀	2 MA 6 MA	KP828476 KP828477	KP828540 KP828541	7 3	26 27
Portugal	Alto Douro	Taomate Freixo de Espada à Cinta Cheires	29 15 16	CDC CDC CDC	♀ ♂	9 MA 5 PL 7PL 8 PL 11 PL	KP828478 KP828450 KP828449 KP828451 KP828452	KP828542 KP828511 KP828510 KP828512 KP828513	16 2 1 4 5	28 2 1 3 2
Portugal (Atlantic Islands)	Savage Islands	Selvagem Grande Island	17	CDC	♂	SSPM1	failed amplification	KP828515	-	5
Spain	Andalousia	Riotinto Torvizcon	18 19	SP CDC	♂	SSPM3 SSPF3 SSPF2	failed amplification failed amplification KP828484	KP828514 KP828516 failed amplification	- - 20	4 5 5
					♂	Esp2 Esp3 4 Esp 6 Esp 10 Esp	KP828453 KP828454 KP828455 KP828456 KP828457	KP828517 KP828518 KP828519 KP828520 KP828521	6 7 8 9 10	6 7 8 9 10
Tunisia	Medlaoui	Medlaoui	5	SP	♂	12 Esp SIPM11 SIPM15 SIPM13 SIPM16	KP828458 KP828493 KP828495 KP828494 KP828491	KP828522 KP828523 KP828556 KP828558 KP828557	4 3 3 28 27	11 11 41 41 39
					♀	S12PM39 S11PF14 S14PF1	KP828492 KP828497 KP828496	KP828555 KP828560 KP828559	7 29 3	40 42 42
Monastir	Monastir	Monastir	6	CDC	♀	SMO 22	KP828498	KP828561	3	43
					♂	SMO 228 SMO 587	KP828499 KP828500	KP828562 KP828563	31 30	44 45

SP, sticky papers; CDC, miniature light traps; UV, ultraviolet CDC miniature light trap.

For the model-based approach, the best-fit model of nucleotide substitutions was computed with JModelTest v2.1.4 (Darriba et al. 2012) using the Akaike information criterion. The best-fit model was the GTR + Γ + I (general time-reversible Model) for the cyt b/COI combined. Bayesian analyses were carried out using MrBayes 3.1.2 (Ronquist and Huelsenbeck 2003) with 2,000,000 generations; 5,000 of the saved trees were discarded and the 15,000 remaining were used to construct the resulting BI tree. One partition for each codon position of both mitochondrial genes was implemented. The robustness of tree nodes was assessed by clade posterior probabilities values.

A specimen of *Sergentomyia schwetzi* Adler, Theodor & Parrot was also processed as an outgroup.

Geographic Distribution of *Se. minuta* Haplotypes and Phylogenetic Networks. Values of haplotype and nucleotide diversity for each marker and each population were analyzed in DnaSP Version 5.10.01 (Librado and Rozas 2009) and their geographic distribution mapped. DNA haplotypes were submitted to Genbank (for accession numbers see Table 1). Analyses of molecular variance (Excoffier et al. 1992), exact tests of population differentiation, and calculation of F -statistic values were done using Arlequin 3.1 (Schneider et al. 2005, Excoffier et al. 2009). ϕ_{ST} values are based on haplotype frequency and molecular distances between haplotypes. Their P-values were calculated from permutation tests (10,000 permutation rounds). To obtain the minimum number of mutations between haplotypes, a network was established with the software Network 4.6.1.3 (www.fluxus-engineering.com; Bandelt et al. 1999) using a maximum parsimony method based on a median joining algorithm.

Molecular Clock. An estimation of the molecular clock has been done by calculation of Cytb pairwise distances and calculations according to the calibration of two hypotheses proposed by Essegir et al. (1997) for Cytb in *Phlebotomus* and *Larroussius* subgenera: 1% variability on the one hand, and 2.5% on the other hand per million years, providing one minimum and one maximum value. This ratio has been applied to the divergences observed in the present study.

Results

COI and Cytb Sequences of *Se. minuta*. Partial COI and Cytb sequences (567 bp and 421 bp without primers, respectively) were obtained for *Se. minuta* from 29 sites in different localities (Table 1). Nucleotide sequences were translated into amino acid sequences according to the invertebrate mitochondrial code; translation yielded protein sequences that contained no stop codons. The sequences of *Se. minuta* from 13 localities contained 40 haplotypes defined by 44 polymorphic sites for Cytb and 53 haplotypes defined by 78 polymorphic sites for COI (Table 1).

For the analyses with Cytb, of the four specimens available from the Savage Islands (Alves-Pires et al. 1997), only one (SSPF2) was successfully amplified and sequenced. The number of base substitutions per site

from averaging over all *Se. minuta* sequence pairs was 2.3%, based on the pairwise analysis of the sequences. A comparison analysis carried out specimen per specimen showed up to 4% variability within *Se. minuta*. For all substitutions, two variants occurred at each polymorphic site. Fifteen substitutions were A $\leftrightarrow$ G transitions, 25 were C $\leftrightarrow$ T transitions, two were A $\leftrightarrow$ T transversion, one T $\leftrightarrow$ G transversion, and one A $\leftrightarrow$ C transversion.

In the analyses from COI, only three specimens (SSPM1, SSPM3, and SSPF3) from the four specimens available from the Savage Islands (Alves-Pires et al. 1997) were successfully amplified and sequenced. The COI matrix includes 78 variables. The number of base substitutions per site from averaging over all *Se. minuta* sequence pairs was 3.3%, based on the pairwise analysis of the sequences. It went up to 5% for several specimens. The mean nucleotide composition in *Se. minuta* was as follow: T: 37.8%, C: 18.5%, A: 28.8%, and G: 15%.

Phylogeny of *Se. minuta* Populations. We used the Cytb and COI concatenated sequences of *Se. minuta* to construct phylogenetic trees. Maximum likelihood (data not shown) and Bayesian trees both contained four separate clades and specimens not positioned in any clade (Fig. 1). Each clade was supported by high bootstrap values (63–99) and most of them represented individuals from the same geographic area.

The first lineage included all specimens from North Africa (Tunisia, Algeria, and Morocco; green color-code).

The second lineage included a few specimens from the south of France, Spain, and 8PL from Portugal (red color-code).

The third lineage included many specimens from southern France, all the specimens from Corsica, one from Spain (ESP2), and all specimens from Portugal except 8PL (blue color-code).

The fourth lineage included all the specimens from the Balkans (Greece, Montenegro), Crete, Malta, Cyprus, and the Savage Islands (orange color-code).

The position of specimens SMO587, SMO222, S1PM16, S12PM39 (Tunisia), BEN01, SM561, SM515, SM605 (Algeria), 2MAR, 9MAR (Morocco), and some of those from Spain (ESP3, 4ESP, and 6ESP) are not resolved.

The Bayesian tree built on COI sequences is similar to those obtained by Cytb analysis and concatenated Cytb-COI analyses (data not shown). The main differences observed are the grouping of all specimens from North Africa and some from Spain (ESP3, 4ESP, and 6ESP) in the same clade.

Phylogeographic Patterns and Matrilineal Structure of *Se. minuta* Based on Cytb Analyses. Among the 14 identified haplotypes, 33 were very geographically restricted in locality (unique to each country or area; Fig. 2). Only seven haplotypes were shared between populations, some haplotypes each covering a different large geographic area. The samples from Corsica and some samples from France shared the same haplotype (H11); three North African countries shared haplotype H3; haplotype H4 was

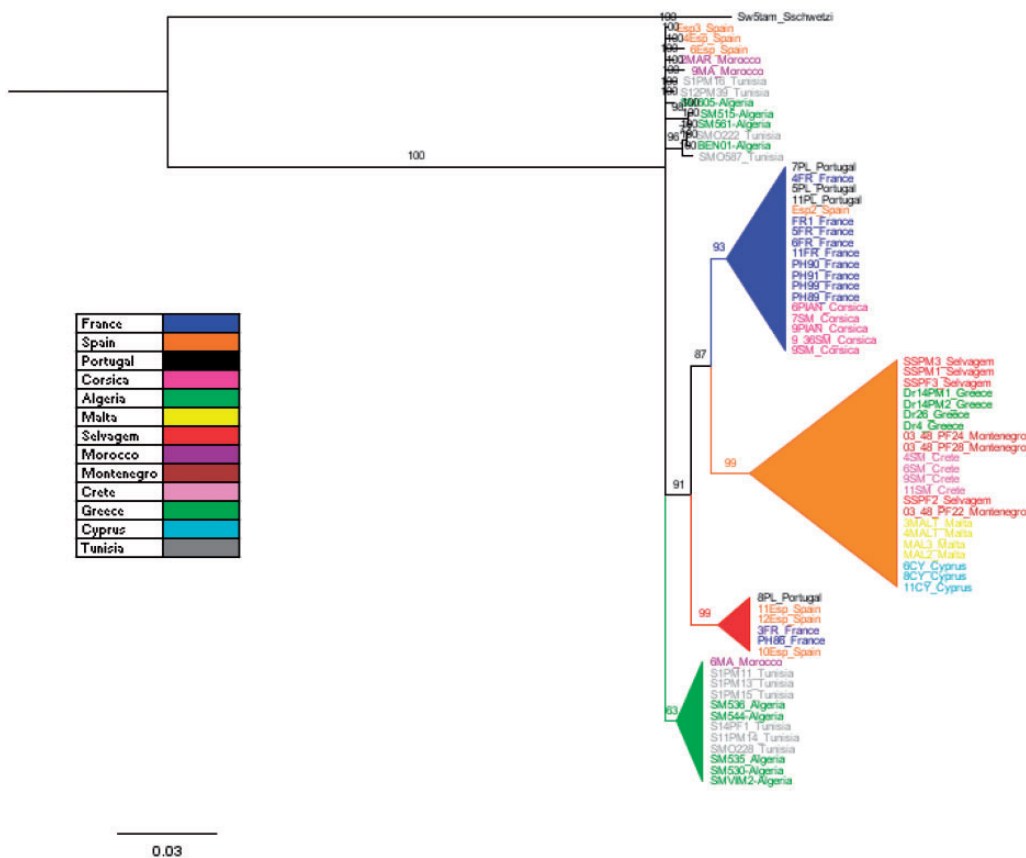


Fig. 1. Bayesian tree resulting from the phylogenetic analysis of the concatenated dataset. Robustness of nodes was indicated by the clade posterior probability values (%). The voucher numbers are followed by the geographic origin. Localities are color-coded.

shared by samples from Spain, France, and Portugal; and haplotype H10 was shared only by samples from France and Spain. Haplotypes H29 and H30 were shared by samples from Algeria and Tunisia, respectively. Samples from Spain shared haplotype H7 with other samples from Algeria and Tunisia. Overall, the distribution of haplotypes in the network (Fig. 2) showed spatial genetic structuring into regional groups: the North African region (13 exclusive haplotypes), European area (11 exclusive haplotypes), and Balkans, Crete, Malta, Cyprus area (14 exclusive haplotypes). Among haplotypes from the European area, haplotypes H4 and H5 were grouped separately in the network (red circles). Samples from Savage Islands (H20) were grouped with samples from the Balkans, Crete, Malta, and Cyprus (Fig. 2).

The samples from the Balkans and those from Crete, Malta, and Cyprus had a haplotype diversity average of 0.971 and a nucleotide diversity average of 0.023. Nucleotide diversity for all samples ranged from 0.001 to 0.026. Haplotype diversity ranged from 0.400 to 1.000. According to the analyses of molecular variance, most of the variance (42.97%) was due to the variation between populations from the three regional groups: Balkans with Crete, Malta, and Cyprus; Europe

(France, Portugal, and Spain); and North Africa. The proportion of molecular variance from populations within each group was 32.42% and was 24.61% between individuals from each population. The corresponding values of ϕ_{sc} , ϕ_{st} , and ϕ_{ct} were 0.431, 0.675, and 0.429, respectively, and these values were significantly different from zero (permutation test, $P < 0.001$; Table 2).

Results from Network and Bayesian analyses were congruent except for the samples 11PL from Portugal and Esp10 from Spain that were swapped between the two analyses.

Discussion

Se. minuta was first described in the area of Parma, Italy, as *Phlebotomus minutus* by Rondani (1843). Newstead (1911) found this species in Malta, and it became the head of a group of sand flies distributed through Europe, Asia, and Africa. This systematic status was then discussed after the discovery of new varieties of this species. In 1948, Theodor placed it in the genus *Sergentomyia*. In 1958, Theodor considered that *Se. minuta* included two subspecies: *Se. minuta minuta* in Europe, having about 40 cibarial teeth, and *Se.*

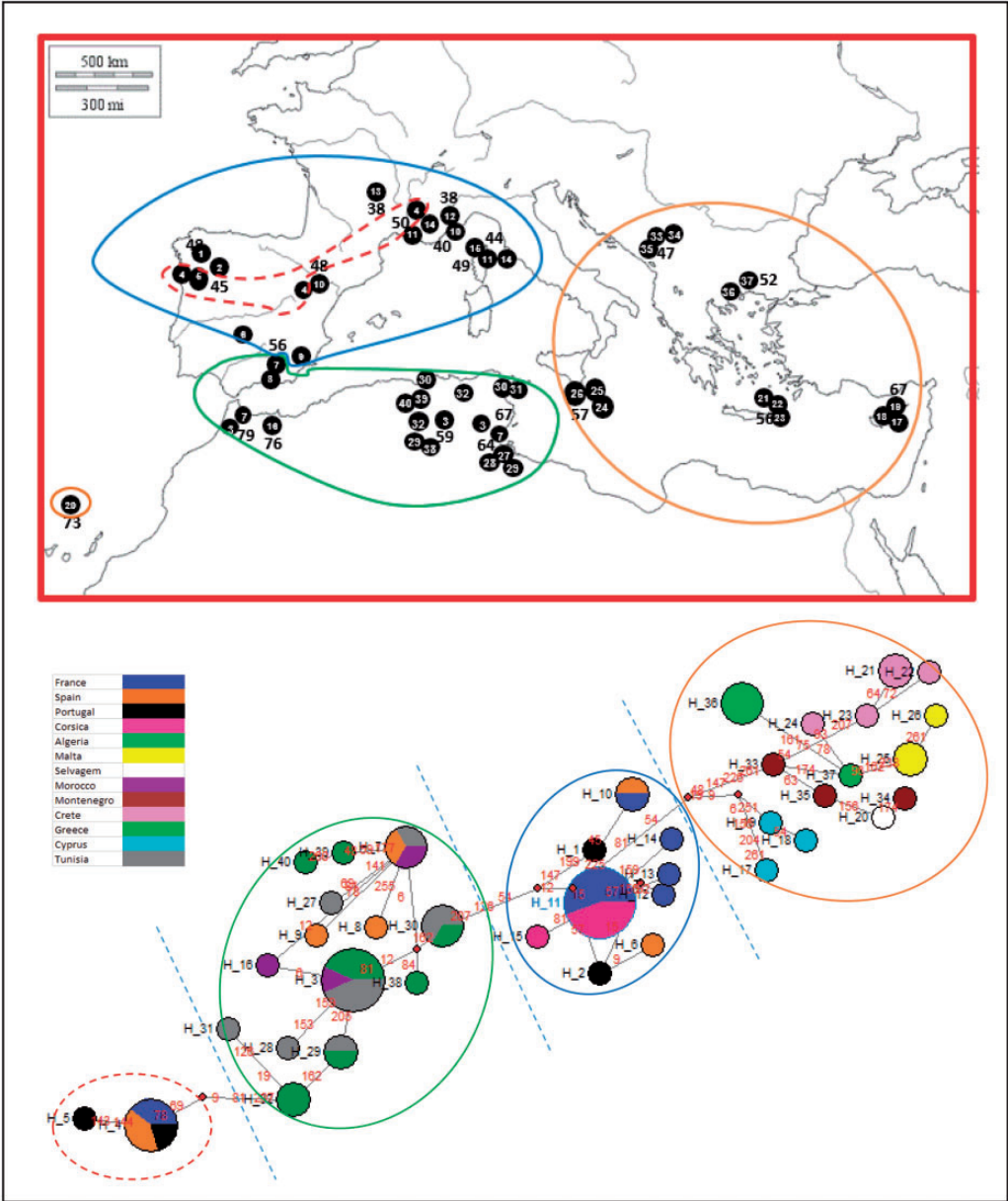


Fig. 2. Geographic distribution of *Se. minuta* haplotypes and generic haplotype networks based on Cytb mtDNA. Hatched line: haplotype distribution range. Sizes of circles are proportional to sample sizes at each locality in the geographic maps and proportional to haplotype frequency in the haplotype network. Haplotypes are color-coded according to their geographic localities. Stippled lines throughout the networks delineate groups of haplotypes according to the population in which they are usually found, but haplotypes can also be shared among populations. Haplotypes are represented in the map by black circles number-coded. Red numbers along branches are IDs of mutations; mv indicates median vectors.

Table 2. Analyses of molecular variance for <i>Se. minuta</i> populations					
Source of variation	Degree of freedom (d.f)	Variance	%	P-value	F-statistic
Among groups of three areas (Balkans with Crete, Malta, and Cyprus, Europe, North Africa)	3	2,323	42.97	0.0000	$F_{sc} = 0.4315$
Among populations within groups	9	1.330	24.61	0.0000	$F_{st} = 0.6758$
Within populations	48	1.752	32.42	0.0003	$F_{ct} = 0.4297$

minuta parroti in North Africa, having about 70 cibarial teeth.

Several authors, mainly in North Africa (Addadi and Dedet 1977, Belazzoug et al. 1982), considered them as two distinct subspecies based on the number of cibarial teeth in the females, whereas others, according to several studies of morphological systematics conducted in Europe and Africa, thought that this character represented a continuous North–South variation (Rioux et al. 1975, Léger et al. 1979, Belazzoug et al. 1982, Pesson et al. 1984). The low number of cibarial teeth observed in European populations (35–60) compared with the high number of teeth observed in North African populations (70–80) would represent the extreme values of its geographical range.

Isoenzyme analysis of *Se. minuta* led Boussa et al. (2009) to conclude that both morphological and isoenzymatic results did not agree with the sub-specific taxonomic rank. They also considered that the genetic structure of *Se. minuta* in the Mediterranean basin needed further studies, especially by inclusion of other populations.

Since the 1990s there has been an increasing interest in comparing the phylogeography between populations of different species of Mediterranean *Phlebotomus* by analyzing the sequences of various genes such as the ribosomal ITS2 or the mitochondrial markers. Studies have shown that *Ph. papatasi*, the main vector of *L. major*, is characterized by a simple genetic structure (Esseghir et al. 1997; Hamarsheh et al. 2007, 2009; Depaquit et al. 2008), whereas *Ph. (Paraphlebotomus) sergenti* Parrot, the main vector of *Leishmania tropica* Wright, is characterized by its high diversity, and a structure highlighting at least four Spanish and Moroccan mitochondrial lineages, which could affect their vectorial capacity (Yahia et al. 2004, Barón et al. 2008). *Ph. (Larroussius) perniciosus* Newstead, one of the main vectors of *L. infantum*, was found by Esseghir et al. (1997) and Pesson et al. (2004) to have several haplotypes (<10). The haplotypes were distributed in two mitochondrial lineages, “the typical lineage,” found in Morocco, Tunisia, Malta, and Italy, and “the Iberian lineage,” found in Spain and also in France (Aransay et al. 2003, Perrotey et al. 2005). These authors suggested that during the Pleistocene ice ages, *Ph. perniciosus* survived in Europe in refugia in southern Spain and Italy, from where there would have been a post-glacial dispersion. *Ph. ariasi* Tonnoir is an important vector of *L. infantum* in some foci located in the western part of the Mediterranean basin. Franco et al. (2012) reported a high intraspecific diversity of 45 haplotypes in 133 specimens that differed from each other by 1–26 mutations. These haplotypes are distributed in two mitochondrial lineages, one of them limited geographically to Algeria and the other widely dispersed across the rest of the Mediterranean countries. Mahamdallie et al. (2010) identified 94 Cytb haplotypes in 452 specimens distributed in five haplogroups: a sublineage suggested the presence of a *Ph. ariasi* refuge north of the Pyrenees during the last glaciation.

The genetic structure of *Se. minuta* has been reported to differ from that of *Ph. perniciosus* by only a few haplotypes despite significant sampling (Pesson et al. 2004, Perrotey et al. 2005, Peyrefitte et al. 2013). Regarding the very high number of haplotypes observed ($n = 40$), the genetic structure of *Se. minuta* is most closely related to that of *Ph. ariasi* (Franco et al. 2012).

The orange lineage includes populations from the Balkans (Greek Macedonia and Montenegro), Crete, Malta, and the Savage Islands. This grouping is strongly supported by bootstrap after rooting on the outgroup (data not shown). A similar grouping (Greek Macedonia, Crete, and Malta) was previously observed for *Ph. similis* after analysis of ITS2 rDNA sequences (Depaquit et al. 2002). At that time, the presence of *Ph. similis* was difficult to explain in Malta because *Ph. sergenti* (and not *Ph. similis*) was recorded in Sicily (D’Urso et al. 2004). Yet, it is interesting to find again a similar grouping with the orange lineage of *Se. minuta*, which implies two different establishments in Malta and in neighboring areas. Should a human importation be considered in Malta? The significant variability in Cytb sequences among populations from Crete, continental Greece, and Malta would suggest that this is not a recent importation by human transport.

Specimens from North Africa are grouped on one branch, which also includes some Spanish specimens from south Andalusia (4ESP, 6ESP, ESP3) without any country individualization, confirming gene flow in North Africa and Spain previously suggested by Boussaa et al. (2009) and in agreement with the status of a unique species in *Se. minuta*. A similar observation was made for *Ph. sergenti* (Barón et al. 2008) and for *Ph. perniciosus* (Pesson et al. 2004, Perrotey et al. 2005, Yahia et al. 2004). Interestingly, a Spanish specimen (ESP2) caught in Rio Tinto, an area having a humid Mediterranean bioclimate, is linked with specimens from Portugal, France, and Corsica, not with other Spanish specimens.

The level of matrilineal subdivision in *Se. minuta* between the eastern (Balkans, Malta, Crete, and Cyprus) and western (South Europe and North Africa) ends of the Mediterranean basin is low. The lack of sharing of Cytb haplotypes between samples from these areas suggests that matrilineal gene flow between these areas is now limited or nonexistent (Fig. 2), suggesting a possible speciation event. Our results suggest that the settlement of the Mediterranean basin could have occurred at the same time for *Phlebotomus* and *Se. minuta*. However, the genetic structure of *Se. minuta* should be completed by 1) sequencing of a non mitochondrial molecular marker and 2) including different specimens and species from other Atlantic islands such as Madeira and the Canary Islands (where sand flies have been recorded) and maybe in Cape Verde and the Azores (where sand flies have never been recorded to our knowledge) to understand their settlement by Phlebotomine sand flies.

According to the speciation events, Esseghir et al. (1997) calibrated the molecular clock of Cytb in *Phlebotomus* and *Larroussius* at 1–2.5% variability per million

years. Considering the variability observed, the divergence time with the most closely related specimens (03-48 PF22 from Montenegro) is estimated to be between 160,000 and 400,000 years whereas the divergence time estimated between the specimen from Savage Islands and those from Continental Greece, Crete, Montenegro, and Malta (differing by 1.5–2%) is estimated to be between 600,000 to 2 million yr. This estimation is not linked to a colonization by Man, taking into consideration that the Savage Islands were discovered in 1438, except if it occurred at a prehistoric time. However, no generalized track could explain the settlement of these islands. Counting revealed a high number of female cibarial teeth (72–76), close to those counted in North Africa. Even if we have not succeeded in sequencing both markers for each specimen, the COI sequences show that two specimens from Savage Islands (SSPF3 and SSPM1) have a position on the tree comparable with that observed by Cytb sequence analysis, whereas SSPM3 has a strongly different haplotype, suggesting two different periods of settlement.

Waiting future studies, we consider the observed variability at the intraspecific level and promote the use of *Se. minuta* all over its distribution area.

Acknowledgments

This work was funded by the project @ SPEED-ID “Accurate SPECiEs Delimitation and Identification of eukaryotic biodiversity using DNA markers” proposed by F-BoL, the French Barcode of life initiative and by The ANSES (PNR EST-11-154).

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Received 9 November 2012; accepted 1 June 2015.